



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

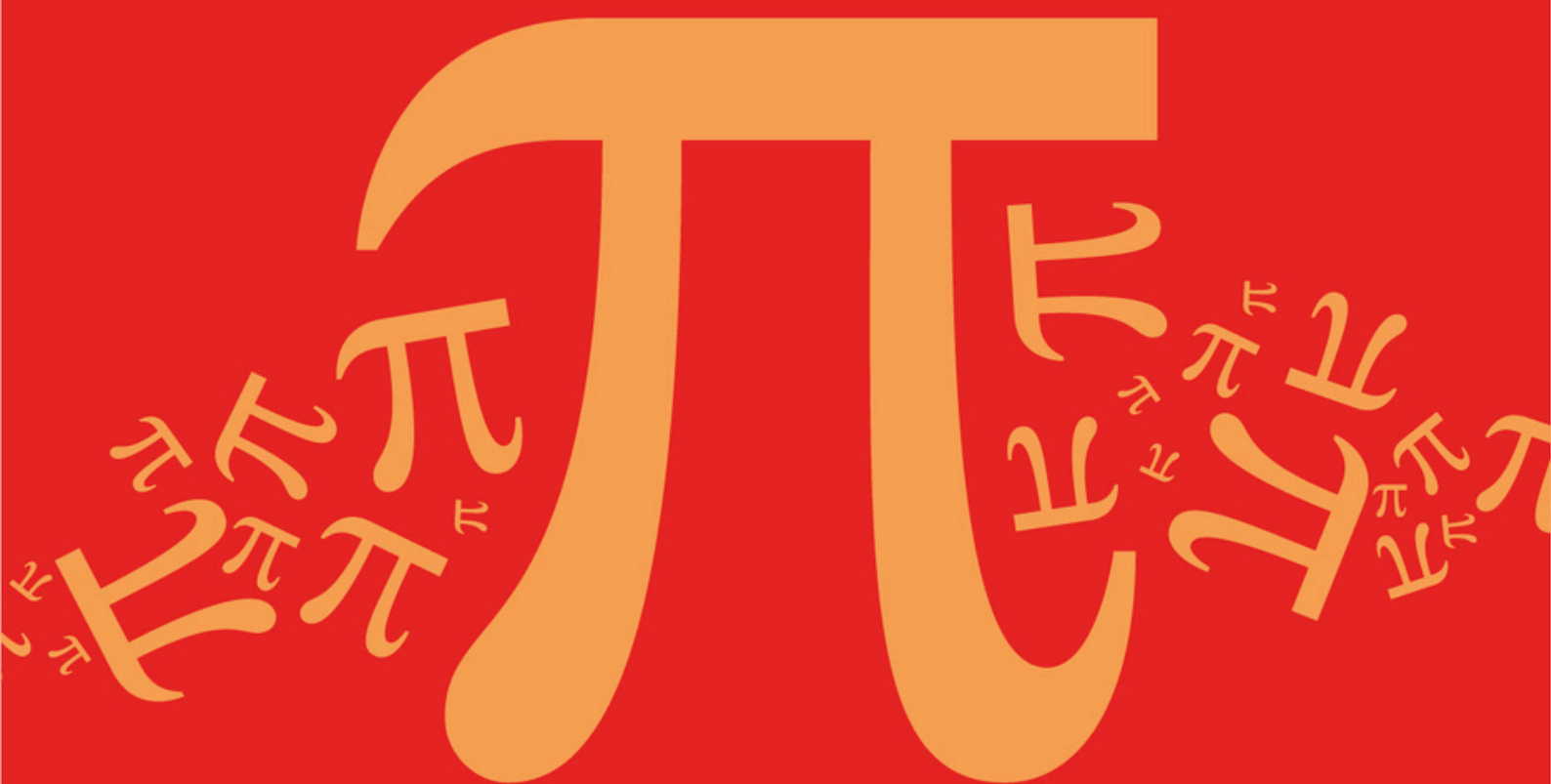
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D10: Modulus Functions and Inequalities

Mark Scheme

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D10: Modulus Functions and Inequalities

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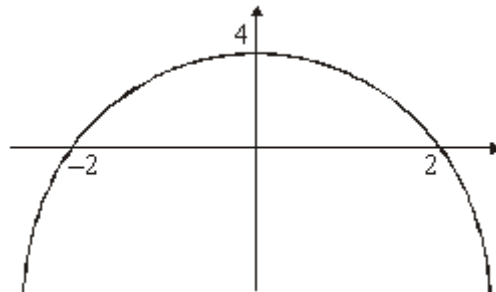
**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) (i)



Shape symmetrical about y axis

all correct

M1

A1

2

(ii) $V = (k) \int (4 - x^2)^2 (dx)$
 $= (\pi) \int 16 - 8x^2 + x^4 dx$
expanding bracket

B1

$= (\pi) \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]$
correctly integrating 2 of their terms

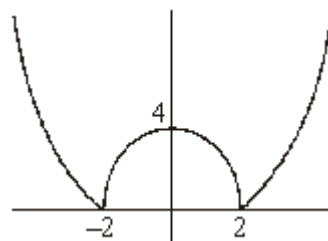
M1

$= \pi \frac{256}{15}$

A1

4

(b) (i)



modulus graph

M1

shape

A1

2

(ii) $|4 - x^2| = 3$

attempt at solving a correct equation

M1

$$4 - x^2 = 3 \Rightarrow x = +1, -1$$

2 correct

A1

$$4 - x^2 = -3 \Rightarrow x = \pm \sqrt{7} \text{ (or exact equivalent)}$$

2 correct

A1

3

(iii) $-\sqrt{7} < x < -1$

condone $\sqrt{7} = 2.6$ (or better)

B1F

$$1 < x < \sqrt{7}$$

B1F

2

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Examiner reports

Q1.

Part (a)(i) was well answered although some scripts used an approach of making a table of values and drawing a graph on graph paper. Part (a)(ii) tested the knowledge and use of the formula for volume of rotation. A number of candidates did not know the correct formula and a significant number of candidates could not expand $(4 - x^2)^2$ correctly. Candidates could draw a modulus graph in part (b), but many had the wrong shape of the curve. Candidates attempted to solve the equation in part (b)(ii) with varying degrees of success. In the final part (b)(iii), successful candidates used the previous graph together with the solution of the intersection from part (b)(ii).

This question was the least well answered question on the paper with very few correct solutions. Although the graphs of inverse trigonometrical functions is stated in the specification there were many incorrect graphs sketched. Many candidates had their calculators in degree mode and in part (b)(ii) found the values of 19 and 21 and then stated, 'change of sign, therefore root'. There were similar mistakes in part (c) with iterations jumping from 0.8 to 44 with candidates apparently not realising there was something amiss.